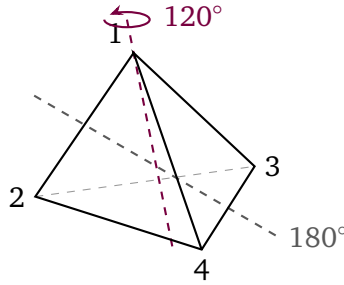


Symmetries in Physics — Exercise Sheet 5

Characters, tables, and projectors

Variant: Questions only

Exercise 1. Construct the character table of A_4 , the rotation group of the tetrahedron. The picture, taken from Lecture 2, shows one rotation axis of each kind:



The coloured axis passes through vertex 1 and the centre of the opposite face. The rotations about it through $\pm 120^\circ$ permute the vertices 2, 3, 4 cyclically. There are four such axes, one through each vertex, which gives eight 3-cycles. The grey axis passes through the midpoints of edges 12 and 34, and the half-turn about it is $(1\ 2)(3\ 4)$. There are three such axes, one for each pair of opposite edges. With the identity, that makes $1 + 3 + 8 = 12$ rotations. Here is everything you will need (Lecture 2's example $A_4/V \cong C_3$).

- $|A_4| = 12$, and A_4 has four conjugacy classes: the identity; the three half-turns $(1\ 2)(3\ 4)$, $(1\ 3)(2\ 4)$, $(1\ 4)(2\ 3)$; the four 3-cycles $(1\ 2\ 3)$, $(1\ 3\ 4)$, $(1\ 4\ 2)$, $(2\ 4\ 3)$; and the four inverse 3-cycles $(1\ 3\ 2)$, $(1\ 4\ 3)$, $(1\ 2\ 4)$, $(2\ 3\ 4)$.
 - The identity and the three half-turns form a normal subgroup V , and $A_4/V \cong C_3$. The projection $\pi: A_4 \rightarrow C_3$ sends V to e , the class of 3-cycles containing $(1\ 2\ 3)$ to r , and the class of $(1\ 3\ 2)$ to r^2 .
 - The characters of C_3 are $\chi_k(r^m) = \omega^{km}$, with $\omega = e^{2\pi i/3}$ and $k = 0, 1, 2$.
- (i) Without constructing any representation, use the two counting theorems of Lecture 5 to find the dimensions of the irreducible representations of A_4 . Do the same for D_6 , which has order 12 and six conjugacy classes (Sheet 2, Exercise 2).
- (ii) Construct three one-dimensional representations of A_4 as $\chi_k \circ \pi$.
- (iii) Find the missing row of the table from $\sum_i n_i \chi_i(g) = 0$ for $g \neq e$, and check its norm.

Exercise 2. Lecture 5's example on D_4 claimed that the quaternion group Q_8 has the same character table. Prove it. Here is everything you will need (Sheet 4, Exercise 6).

- $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, with $i^2 = j^2 = k^2 = ijk = -1$. Any two of i, j, k anticommute, for instance $ij = k = -ji$.
- Four one-dimensional representations ψ_{ab} , with $a, b = \pm 1$:

$$\psi_{ab}(\pm 1) = 1, \quad \psi_{ab}(\pm i) = a, \quad \psi_{ab}(\pm j) = b, \quad \psi_{ab}(\pm k) = ab.$$

- A two-dimensional irreducible representation

$$i \mapsto -i\sigma_1, \quad j \mapsto -i\sigma_2, \quad k \mapsto -i\sigma_3, \quad -1 \mapsto -\mathbb{1}.$$

- The table of D_4 (Lecture 5):

D_4	$1\ e$	$1\ r^2$	$2\ r$	$2\ s$	$2\ rs$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

The lecture labels these rows $\chi_1, \dots, \chi_5$; here we use Mulliken's symbols.

- Find the conjugacy classes of Q_8 and their sizes.
- Prove that these five representations are all the irreducible representations of Q_8 .
- Write down the character table of Q_8 and compare it with that of D_4 .
- Prove that $D_4 \not\cong Q_8$. What does this say about the character table?

Exercise 3. S_4 permutes the coordinates of $\mathbb{C}^4$. Compute the character of this permutation representation, then use the character calculus to decompose it and to prove that its three-dimensional summand is irreducible.

Exercise 4. Let ρ be a d -dimensional representation of a finite group G , and χ its character.

- Show that $|\chi(g)| \leq d$ for every g , with equality if and only if $\rho(g)$ is a scalar multiple of the identity. Deduce that $\chi(g) = d$ exactly on the kernel of ρ .
- Suppose that ρ is faithful. Prove that $|\chi(g)| = d$ only for central elements g .
- From the tables of Exercise 2, find which irreducible representations of D_4 and Q_8 are faithful, and at which elements $|\chi(g)| = d$.

Exercise 5. This exercise and the next use Lecture 6's character projectors. In Lecture 6's example of a particle hopping on the corners of a triangle, $C_3 = \langle r \rangle$ acts on the orbitals by

$$\rho(r) |j\rangle = |j+1\rangle,$$

with indices taken modulo 3. This is the regular representation of C_3 (Sheet 3, Exercise 5). The lecture simply wrote down the vectors $v_k = \sum_{j=1}^3 \omega^{-kj} |j\rangle$, where $\omega = e^{2\pi i/3}$. Derive them from the character projectors.

- Write down the three character projectors P_k of C_3 .
- Apply each of them to $|1\rangle$, and compare the result with v_k .
- Check that $P_0 + P_1 + P_2 = 1$, and explain why the image of each P_k is a line.

Exercise 6 (retrieval: parity through character projectors). Return to the parity action of $C_2 = \{e, p\}$ on $L^2(\mathbb{R})$ from Sheet 3, Exercise 4: $(\Pi\psi)(x) = \psi(-x)$, where $\Pi = \rho(p)$. Using the two characters and the character-projector formula of Lecture 6, compute both projectors and identify their images. The formula was proved for finite-dimensional spaces, but for a group of order 2 the averaging makes sense on $L^2(\mathbb{R})$ directly, and you may use it in that form. Explain why each image is an isotypic component but is not itself an individual irreducible representation.